JUNIOR SCHOOL LEARNING HUB

GRADE 9 MATHEMATICS
END-OF-TERM 2 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 2	Learning Area: Mathematics

INSTRUCTIONS TO LEARNERS

- Answer all questions in Section A and Section B.
- Show all working clearly. Marks may be awarded for correct method.
- Use π = 22/7 where necessary unless otherwise stated.
- A non-programmable calculator may be used where appropriate.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Simplify: $3(2x - 5) - 2(x + 4)$. (2 marks)

2. Solve the equation $5x - 7 = 3x + 9$. (2 marks)

3. A trader buys an item for KSh 800 and sells it for KSh 920. Calculate the percentage profit. (2 marks)

4. Express 0.00056 in standard form. (2 marks)

5. Factorise completely: $x^2 + 7x + 12$. (2 marks)

6. Find the gradient of the line passing through (2, 3) and (6, 11). (2 marks)

7. The angles of a triangle are x, 2x and 3x. Find the largest angle. (2 marks)

8. Convert 3.6 km into metres. (2 marks)

9. Find the area of a circle of radius 7 cm. Use π = 22/7. (2 marks)

10. A bag contains 5 red, 3 blue and 2 green balls. Find the probability of picking a blue ball. (2 marks)

11. Evaluate $2^3 \times 2^4 \div 2^2$. (2 marks)

12. Find the mean of 6, 8, 10, 12 and 14. (2 marks)

13. Make y the subject of the formula: $x = 3y - 5$. (2 marks)

14. A map scale is 1:50,000. What actual distance is represented by 4 cm? Give your answer in kilometres. (2 marks)

15. Find the circumference of a circle of diameter 14 cm. Use π = 22/7. (2 marks)

16. Solve simultaneously: $x + y = 11$ and $x - y = 3$. (2 marks)

17. A right-angled triangle has perpendicular sides 6 cm and 8 cm. Find its hypotenuse. (2 marks)

18. Write 35% as a fraction in its simplest form. (2 marks)

19. Determine the next two terms in the sequence 4, 9, 14, 19, … (2 marks)

20. A learner scores 18 marks out of 24. Express the score as a percentage. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Algebra (10 marks)

a) Expand and simplify: $(2x - 3)(x + 4)$. (3 marks)

b) Factorise: $2x^2 + 7x + 3$. (3 marks)

c) Hence solve $2x^2 + 7x + 3 = 0$. (4 marks)

22. Linear relationships and graphs (10 marks)

The relationship between x and y is $y = 2x - 1$.

a) Complete the table for $x = -1, 0, 1, 2, 3$. (3 marks)

b) Draw the graph of $y = 2x - 1$ for $-1 \leq x \leq 3$. (4 marks)

c) State the gradient and y-intercept. (3 marks)

23. Geometry and mensuration (10 marks)

A cylindrical water tank has radius 3.5 m and height 4 m. Use $\pi = 22/7$.

a) Calculate the area of its circular base. (3 marks)

b) Calculate its volume. (3 marks)

c) The tank is 75% full. Find the volume of water in it. (4 marks)

24. Statistics (10 marks)

The marks scored by ten learners are: 12, 15, 10, 18, 15, 14, 16, 15, 11, 14.

a) Arrange the data in ascending order. (2 marks)

b) Find the mean. (3 marks)

c) Find the median and mode. (3 marks)

d) State the range. (2 marks)

25. Similarity, scale and right-angled triangles (10 marks)

a) Two similar triangles have corresponding sides in the ratio 2:5. If the shorter corresponding side is 8 cm, find the longer side. (3 marks)

b) A ladder 10 m long rests against a vertical wall. Its foot is 6 m from the wall. How high up the wall does it reach? (4 marks)

c) A drawing uses a scale of 1:100. A wall is 6.5 cm long on the drawing. Find its actual length in metres. (3 marks)

26. Probability and financial mathematics (10 marks)

A school club has 12 boys and 18 girls. A learner is selected at random.

a) Find the probability that the learner selected is a girl. (2 marks)

b) Find the probability that the learner selected is a boy. (2 marks)

c) A shop gives a 10% discount on a bag marked KSh 2,400. Calculate the discount and the selling price. (3 marks)

d) KSh 5,000 is invested at 8% simple interest per annum for 2 years. Find the interest and total amount. (3 marks)

MARKING SCHEME

Grade 9 Mathematics – End-of-Term 2 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. $6x - 23$ (2)

2. $2x = 16$; $x = 8$ (2)

3. Profit = 120; percentage profit = $120/800 \times 100 = 15\%$ (2)

4. 5.6×10^{-4} (2)

5. $(x + 3)(x + 4)$ (2)

6. $(11 - 3)/(6 - 2) = 2$ (2)

7. $6x = 180$; $x = 30°$; largest = $90°$ (2)

8. 3,600 m (2)

9. 154 cm^2 (2)

10. 3/10 (2)

11. $2^5 = 32$ (2)

12. 10 (2)

13. $y = (x + 5)/3$ (2)

14. $4 \times 50{,}000 = 200{,}000$ cm = 2 km (2)

15. $\pi d = 44$ cm (2)

16. $x = 7$, $y = 4$ (2)

17. $\sqrt{(6^2 + 8^2)} = 10$ cm (2)

18. 7/20 (2)

19. 24, 29 (2)

20. $18/24 \times 100 = 75\%$ (2)

SECTION B (60 MARKS)

21. Algebra (10 marks)

$2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$. (3)

$(2x + 1)(x + 3)$. (3)

$2x + 1 = 0$ or $x + 3 = 0$; $x = -1/2$ or $x = -3$. (4)

22. Linear relationships and graphs (10 marks)

y values: −3, −1, 1, 3, 5. (3)

Correctly plotted points and straight line. (4)

Gradient = 2; y-intercept = −1. (3)

23. Geometry and mensuration (10 marks)

$\pi r^2 = 22/7 \times 3.5 \times 3.5 = 38.5\ m^2$. (3)

$38.5 \times 4 = 154\ m^3$. (3)

$0.75 \times 154 = 115.5\ m^3$. (4)

24. Statistics (10 marks)

10, 11, 12, 14, 14, 15, 15, 15, 16, 18. (2)

Total = 140; mean = 140 ÷ 10 = 14. (3)

Median = (14 + 15)/2 = 14.5; mode = 15. (3)

18 − 10 = 8. (2)

25. Similarity, scale and right-angled triangles (10 marks)

Scale factor = 5/2; longer side = 8 × 5/2 = 20 cm. (3)

$Height^2 = 10^2 - 6^2 = 64$; height = 8 m. (4)

6.5 × 100 = 650 cm = 6.5 m. (3)

26. Probability and financial mathematics (10 marks)

18/30 = 3/5. (2)

12/30 = 2/5. (2)

Discount = 0.10 × 2,400 = KSh 240; selling price = KSh 2,160. (3)

I = PRT/100 = 5,000 × 8 × 2/100 = KSh 800; amount = KSh 5,800. (3)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Mathematics teacher should check that the topics assessed match the actual Grade 9 Term 2 teaching sequence and the school scheme of work. Accept mathematically equivalent methods and correct units where appropriate.

Prepared by: Junior School Learning Hub